Cauchy coefficients are Taylor coefficients: the derivative identification (Taylor-tree Stage 8)

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Abstract

Units 266–268 close the last qualification of the Taylor-tree programme: the several-variable Cauchy coefficients $c_\gamma = A_r^{[d]}(F \prod_i w_i^{-\gamma_i})$ of Stage 7 are the normalised Taylor derivatives $\partial^\gamma F(0)/\gamma!$, $\gamma! = \prod_i \gamma_i!$, for F holomorphic on the open polydisc of radius $R > r$. Consequently `thm:TaylorTree` and `cor:standardintegralexp` hold, in every dimension and under the paper's own hypothesis, with the coefficient families $\gamma \mapsto \mathrm{Re}(\partial^\gamma F(0)/\gamma!)$ and the original integral (Headline XXXIII, `thm_TaylorTree_taylor`). Astra #32 capped the sprint at six units with a gate after the first; it closed in three. No `sorry`, no additional `axiom`; 9143 jobs; pin `066b8c6`. The Taylor-tree programme is frozen at this pin.

1 The design (Astra #32)

The obvious route — differentiate the contour integral under the integral sign to arbitrary order — was rejected. Instead the target is the *fixed-order* iterated coordinate derivative

$$\partial^\gamma F(0) = \partial_x^{\gamma_0}\Big[x \mapsto \partial^{\gamma'} F(x,\cdot)(0)\Big](0),$$

tail coordinates first and the head coordinate last (`coordDeriv`), with ordinary `iteratedDeriv` at 0. By the recursion $c_{(\gamma_0,\gamma')}(F) = A_r\big(x \mapsto x^{-\gamma_0} g(x)\big)$ with $g(x) = c_{\gamma'}(F(x,\cdot))$, and $A_r(x^{-n}g)$ is the one-variable Cauchy coefficient of g (`cauchyPowerSeries`), so the one-variable identification of Stage 6 applies provided g is holomorphic on a neighbourhood of the closed disc of radius r. The induction hypothesis then gives $g(x) = \partial^{\gamma'} F(x,\cdot)(0)/\gamma'!$ for every x in the parameter disc, an equality of functions near 0, so the γ_0-th derivative may be taken of the right-hand side (`Filter.EventuallyEq.iteratedDeriv_eq`). The only analytic obligation is the holomorphy of g in the parameter.

2 The gate: holomorphy in a parameter without differentiating integrals (unit 266)

Two identities make this cheap.

- **The circle operator is the circle average.** Since $\partial_\theta(re^{i\theta}) = ire^{i\theta}$, $A_r(g) = (2\pi i)^{-1}\oint w^{-1} g = (2\pi)^{-1}\int_0^{2\pi} g(re^{i\theta})\,d\theta$ (`circleOp_eq_integral`). With this, Fubini for two circle operators with a jointly continuous integrand is `MeasureTheory.integral_integral_swap` on $[0,2\pi]^2$ (`Measure.prod_restrict`, `ContinuousOn.integrableOn_compact` on a product of compact intervals), and induction along `Fin.cons` with the parametric-continuity lemma of unit 262 gives $A_r^{[d]}\big(w \mapsto A_\rho(z \mapsto K(z,w))\big) = A_\rho\big(z \mapsto A_r^{[d]}(K(z,\cdot))\big)$ (`iterOp_circleOp_swap`).

- **Cauchy-type integrals are holomorphic.** Mathlib's `hasFPowerSeriesOn_cauchy_integral` says $x \mapsto (2\pi i)^{-1} \oint_{|z|=\rho} (z-x)^{-1} g(z)\, dz$ has a power series on $|x| < \rho$ for circle-integrable g; rewriting $z^{-1}(1-x/z)^{-1} = (z-x)^{-1}$ gives `differentiableOn_circleOp_cauchyKernel`.

For $r < \rho < R$ and $|x| < \rho$, the slice Cauchy formula $F(x,w') = A_\rho\big(z \mapsto (1-x/z)^{-1} F(z,w')\big)$ holds for every w' on the torus of radius r (unit 260), so

$$g(x) = A_r^{[d]}\Big(w' \mapsto A_\rho\big(z \mapsto (1-x/z)^{-1} F(z,w') \prod w'^{-\gamma'}\big)\Big) = A_\rho\big(z \mapsto (1-x/z)^{-1} g(z)\big),$$

and g is continuous on the circle of radius ρ by parametric continuity. Hence g is holomorphic on $|x| < \rho \supset \{|x| \le r\}$ (`differentiableOn_polyCoeff_param`). The gate passed in one unit, as Astra #32 required.

3 Identification and the paper-facing theorem (units 267–268)

`circleOp_eq_discCoeff` is `cauchyPowerSeries_apply` at $w = 1$; `polyCoeff_eq_discCoeff` is the recursion `polyCoeff_cons` of unit 263. The induction `polyCoeff_eq_coordDeriv_div` takes $\rho = (r+R)/2$, applies `discCoeff_eq_iteratedDeriv_div` (which wants differentiability on the closed disc of radius r as an `NNReal`), rewrites g near 0 by the induction hypothesis, and pulls out the constant $\gamma'!$ with `iteratedDeriv_div_const`; $\gamma! = \gamma_0!\,\gamma'!$ is `Fin.prod_univ_succ`. Unit 268 defines `taylorFamily` $F\,\gamma = \mathrm{Re}(\partial^\gamma F(0)/\gamma!)$, proves it equal to the real Cauchy family of Headline XXXII, and restates `thm_TaylorTree_analytic'` for it (`thm_TaylorTree_taylor`); `taylorFamily` $F\,0 = \mathrm{Re}\,F(0)$.

4 Non-claims

Fixed-order derivatives: no permutation-invariance theorem (for holomorphic F any order gives the usual mixed partial). The families are the Taylor coefficients of the canonical analytic representative $\mathrm{Re}\,F$ at 0, not of the supplied real functions ξ, η, which the hypotheses constrain only on the positive box $(0,b]^d$; when F_ξ is a genuine holomorphic extension of a real-analytic ξ near 0 these are the Taylor coefficients of ξ. Cutoff independence means independence from the spectral threshold L; candidate support is containment, not nonvanishing; no resolution, general-domain, Mellin-continuation or posterior-weak-convergence statement is implied.

5 Lessons

- **Use the induction hypothesis as a local identity.** Once the parametric coefficient is known to be holomorphic, the recursion never needs derivatives of integrals: the derivative is taken of the right-hand side of the induction hypothesis.

- **Normalise contour integrals to averages before Fubini.** The circle-average form removes the Jacobian and makes the double integral a plain product-measure statement.

- **`NNReal` literals in statements.** `(0 : NNReal) < r, h` elaborates the anonymous constructor as a bare subtype and the rewrite fails with a confusing transparency note; bind `set rN : NNReal := r, h` first and rely on the definitional coercion.

- **Deprecations on this pin.** `Metric.emetric_ball_nnreal` is `Metric.eball_coe`; `fun_prop` does not see through `circleMap`, so compose `continuous_circleMap` by hand.

- **A gate that is a lemma, not a feeling.** Astra #32's stop criteria (no arbitrary-order dominated-differentiation hierarchy, no multilinear-series bridge, no strengthening of hypotheses) were checkable at the end of unit 266; the sprint then had a fixed shape.